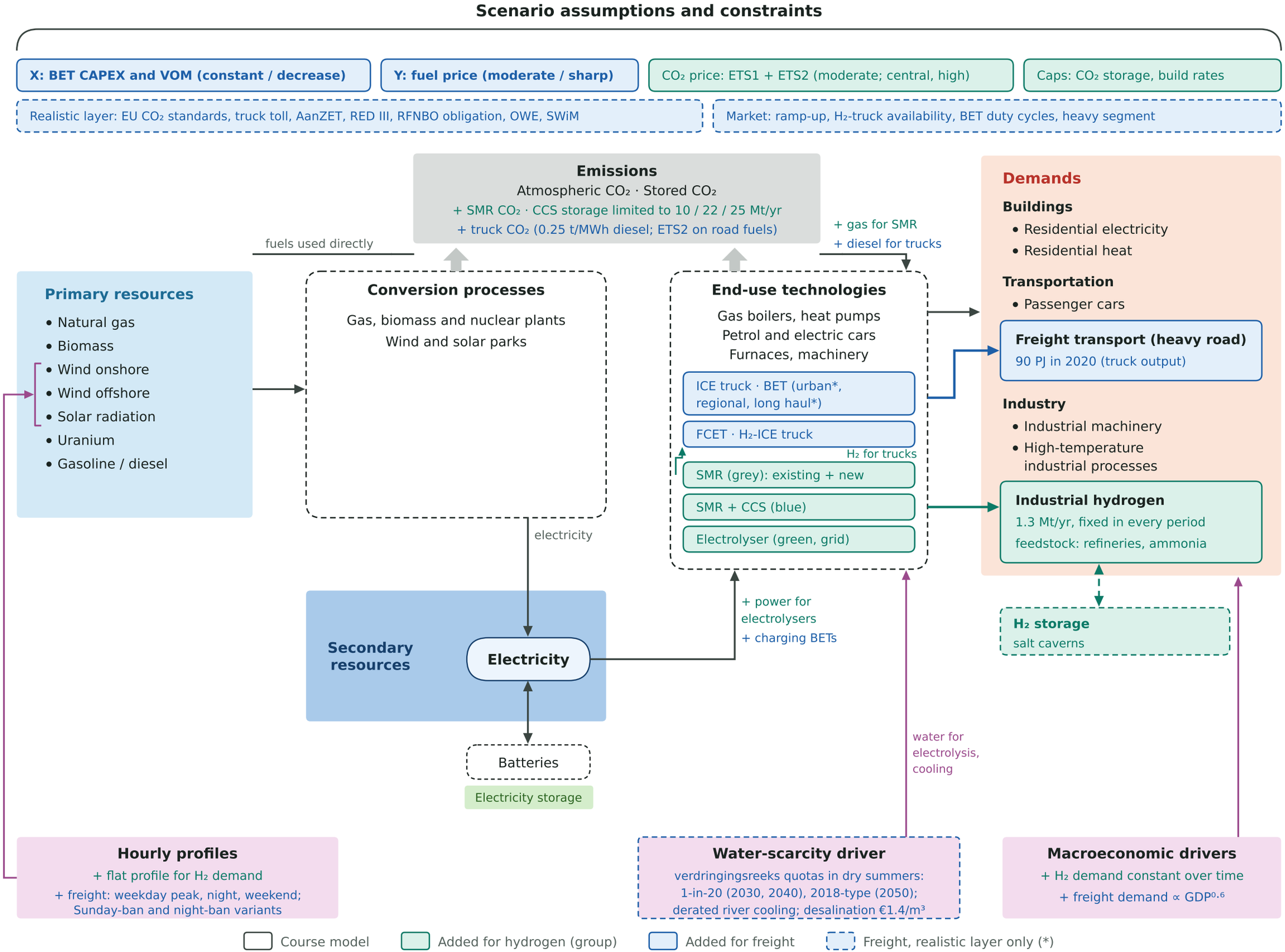
Scenario assumptions and constraints
X: BET CAPEX and VOM (constant / decrease)
Y: fuel price (moderate / sharp)
CO₂ price: ETS1 + ETS2 (moderate; central, high)
Caps: CO₂ storage, build rates
Realistic layer: EU CO₂ standards, truck toll, AanZET, RED III, RFNBO obligation, OWE, SWiM
Market: ramp-up, H₂-truck availability, BET duty cycles, heavy segment
Emissions
Atmospheric CO₂ · Stored CO₂
+ SMR CO₂ · CCS storage limited to 10 / 22 / 25 Mt/yr
+ truck CO₂ (0.25 t/MWh diesel; ETS2 on road fuels)
fuels used directly
+ gas for SMR
+ diesel for trucks
Primary resources
Natural gas
Biomass
Wind onshore
Wind offshore
Solar radiation
Uranium
Gasoline / diesel
Conversion processes
Gas, biomass and nuclear plants
Wind and solar parks
End-use technologies
Gas boilers, heat pumps
Petrol and electric cars
Furnaces, machinery
ICE truck · BET (urban*, regional, long haul*)
FCET · H₂-ICE truck
H₂ for trucks
SMR (grey): existing + new
SMR + CCS (blue)
Electrolyser (green, grid)
Demands
Buildings
Residential electricity
Residential heat
Transportation
Passenger cars
Freight transport (heavy road)
90 PJ in 2020 (truck output)
Industry
Industrial machinery
High-temperature industrial processes
Industrial hydrogen
1.3 Mt/yr, fixed in every period
feedstock: refineries, ammonia
H₂ storage
salt caverns
electricity
Secondary resources
Electricity
+ power for electrolysers
+ charging BETs
Batteries
Electricity storage
water for electrolysis, cooling
Hourly profiles
+ flat profile for H₂ demand
+ freight: weekday peak, night, weekend; Sunday-ban and night-ban variants
Water-scarcity driver
verdringingsreeks quotas in dry summers: 1-in-20 (2030, 2040), 2018-type (2050); derated river cooling; desalination €1.4/m³
Macroeconomic drivers
+ H₂ demand constant over time
+ freight demand ∝ GDP^0.6
Course model
Added for hydrogen (group)
Added for freight
Freight, realistic layer only (*)